

DIGESTIVE SYSTEM

The digestive tube (alimentary canal) is basically a tube where food is taken in, broken down for absorption into the blood stream and the undigested food passed out the other end.

The parts of the system are as follows.

Mouth

- Teeth and tongue work together to cut and grind food into small pieces.
- Saliva contains an enzyme to break down starch.

Gullet

- Also called *oesophagus*.
- Muscles contract and push food down (called *peristalsis*).

Stomach

- Muscles mechanically churn the food and break it down.
- More enzymes are released by glands into the stomach to chemically digest the food.

Small Intestine

- Bile is secreted by the *liver*, stored in the *gall bladder* and released into the *small intestine*.
- *Pancreas* releases a juice containing the three digestive enzymes.

i.e. **Amylase** - digests starch into sugars.
 Protease - digests protein into amino acids.
 Lipase - digests fats and oils.

- *Villi* are finger-like projections on the intestine wall that increases the surface area for absorption of food.

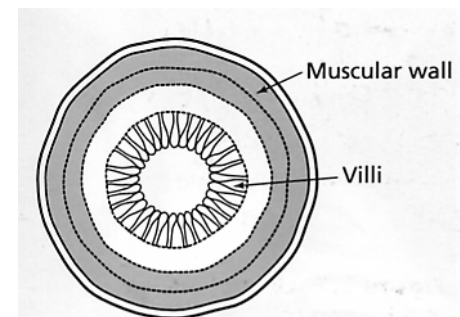
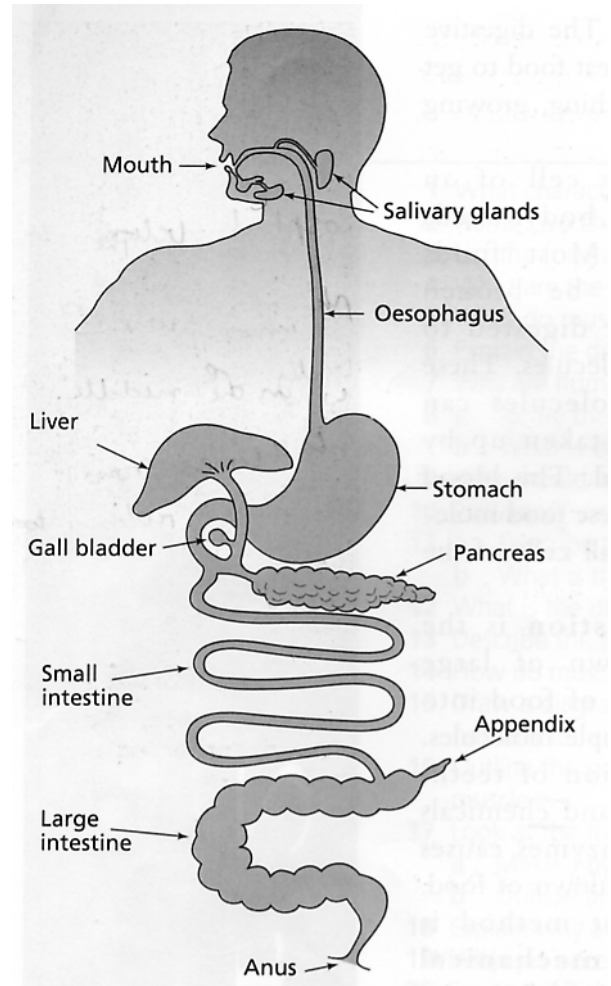


Figure 2.6 Cross-section of the small intestine.

- Food particles move through the wall into the blood stream.

Large Intestine

- Water and dissolved salts are absorbed into the blood stream.

Anus

- Passes undigested food out as *faeces*.

Note

Faeces is not a waste product of the body!

The digestive system is concerned with:

Digestion - breakdown of food by mechanical and chemical means.

Absorption - of the soluble products of digestion through the intestine wall into the blood.

Elimination - removal of undigested food from the body.

Types of Digestion

Mechanical Digestion

- Physical process of breaking food into smaller pieces by the teeth.
- Muscles in the stomach also squeeze food.

Chemical Digestion

- Saliva and stomach juices (enzymes) break large food particles into small molecules that are soluble in water.
- Molecules can pass through the walls of the intestines into the bloodstream.